Center of Mass of a Rod with Quadratic Density

A deliberately step-by-step solution focused on the physics, with integration treated as a calculation tool.

Problem

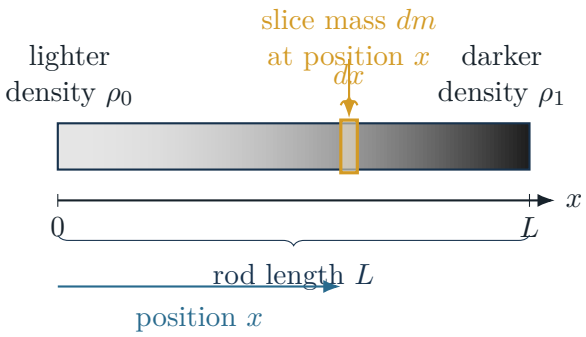
A thin rod of length L lies along the x-axis from $x = 0$ to $x = L$. Its *linear mass density* is

$$\boxed{\rho(x) = \rho_0 + (\rho_1 - \rho_0)\left(\frac{x}{L}\right)^2}.$$

Here ρ_0 is the density at $x = 0$ and ρ_1 is the density at $x = L$. Find the center of mass of the rod in terms of L, ρ_0, and ρ_1.

A physical picture of the rod

The shading below is light at $x = 0$ and becomes darker according to a quadratic rule as x approaches L. Darker shading represents more mass per unit length. (The drawing shows the common case $\rho_1 > \rho_0$; the algebra also covers $\rho_1 < \rho_0$.)



Mass of the highlighted slice. Density means mass per unit length. Therefore

$$\boxed{dm = \rho(x)\,dx = \left[\rho_0 + (\rho_1 - \rho_0)\left(\frac{x}{L}\right)^2\right]dx.}$$

The integral sign will simply add the masses (or moments) of all such slices from $x = 0$ to $x = L$.

1. The center-of-mass recipe

For separate point masses, the one-dimensional center of mass is a mass-weighted average:

$$x_{\mathrm{CM}} = \frac{m_1 x_1 + m_2 x_2 + \cdots}{m_1 + m_2 + \cdots}.$$

A continuous rod follows exactly the same idea, except that it contains very many tiny slices. The denominator is the total mass, while the numerator is the total *moment* (mass times position):

$$x_{\mathrm{CM}} = \frac{\displaystyle\int_0^L x\, dm}{\displaystyle\int_0^L dm} = \frac{\displaystyle\int_0^L x\rho(x)\, dx}{\displaystyle\int_0^L \rho(x)\, dx}.$$

The only integration fact needed:

$$\int_0^L x^n\, dx = \frac{L^{n+1}}{n+1}.$$

In particular,

$$\int_0^L dx = L, \qquad \int_0^L x\, dx = \frac{L^2}{2}, \qquad \int_0^L x^2\, dx = \frac{L^3}{3}, \qquad \int_0^L x^3\, dx = \frac{L^4}{4}.$$

2. Find the total mass M (the denominator)

Begin with $M = \int_0^L \rho(x)\, dx$ and insert the density:

$$\begin{aligned} M &= \int_0^L \left[\rho_0 + (\rho_1 - \rho_0)\left(\frac{x}{L}\right)^2\right] dx \\ &= \rho_0 \int_0^L dx + \frac{\rho_1 - \rho_0}{L^2} \int_0^L x^2\, dx. \end{aligned}$$

Now use $\int_0^L dx = L$ and $\int_0^L x^2\, dx = L^3/3$:

$$\begin{aligned} M &= \rho_0 L + \frac{\rho_1 - \rho_0}{L^2}\left(\frac{L^3}{3}\right) \\ &= \rho_0 L + \frac{(\rho_1 - \rho_0)L}{3} \\ &= \frac{3\rho_0 L + \rho_1 L - \rho_0 L}{3} \\ &= \boxed{\frac{L(2\rho_0 + \rho_1)}{3}}. \end{aligned}$$

This result is reasonable dimensionally: density has units of mass/length, so $L\rho$ has units of mass.

3. Find the total moment N (the numerator)

To keep the algebra readable, call the numerator N:

$$\begin{aligned} N &= \int_0^L x\rho(x)\,dx \\ &= \int_0^L x\left[\rho_0 + (\rho_1 - \rho_0)\left(\frac{x}{L}\right)^2\right] dx \\ &= \rho_0 \int_0^L x\,dx + \frac{\rho_1 - \rho_0}{L^2}\int_0^L x^3\,dx. \end{aligned}$$

Use $\int_0^L x\,dx = L^2/2$ and $\int_0^L x^3\,dx = L^4/4$:

$$\begin{aligned} N &= \rho_0\frac{L^2}{2} + \frac{\rho_1 - \rho_0}{L^2}\left(\frac{L^4}{4}\right) \\ &= \frac{\rho_0 L^2}{2} + \frac{(\rho_1 - \rho_0)L^2}{4} \\ &= \frac{2\rho_0 L^2 + \rho_1 L^2 - \rho_0 L^2}{4} \\ &= \boxed{\frac{L^2(\rho_0 + \rho_1)}{4}}. \end{aligned}$$

4. Divide moment by mass

Now substitute N and M into $x_{\mathrm{CM}} = N/M$:

$$\begin{aligned} x_{\mathrm{CM}} &= \frac{\dfrac{L^2(\rho_0 + \rho_1)}{4}}{\dfrac{L(2\rho_0 + \rho_1)}{3}} \\ &= \frac{L^2(\rho_0 + \rho_1)}{4}\,\frac{3}{L(2\rho_0 + \rho_1)} \\ &= \boxed{\frac{3L(\rho_0 + \rho_1)}{4(2\rho_0 + \rho_1)}}. \end{aligned}$$

Final answer

$$\boxed{\mathbf{r}_{\mathrm{CM}} = \left(\frac{3L(\rho_0 + \rho_1)}{4(2\rho_0 + \rho_1)},\, 0,\, 0\right)}.$$

The y- and z-coordinates vanish because the entire rod lies on the x-axis.

5. Checks that make physical sense

Situation	Formula gives	Interpretation
$\rho_1 = \rho_0$	$x_{\text{CM}} = L/2$	Uniform rod: the center is the midpoint.
$\rho_0 = 0$	$x_{\text{CM}} = 3L/4$	Density grows toward $x = L$, so the center shifts right.
$\rho_1 = 0$	$x_{\text{CM}} = 3L/8$	Density decreases toward $x = L$, so the center shifts left.

Units check. The density factors cancel in the ratio, leaving one factor of L. A center-of-mass coordinate must have units of length, so the result is dimensionally consistent.

6. Three specific density ratios

We now apply the general result

$$x_{\mathrm{CM}} = \frac{3L(\rho_0 + \rho_1)}{4(2\rho_0 + \rho_1)}$$

to three increasingly nonuniform rods. In each drawing, the gold marker shows the calculated center of mass. The midpoint $L/2$ is included as a dashed reference line so that the shift is easy to see.

Case 1: $\rho_1 = \rho_0$ (uniform density)

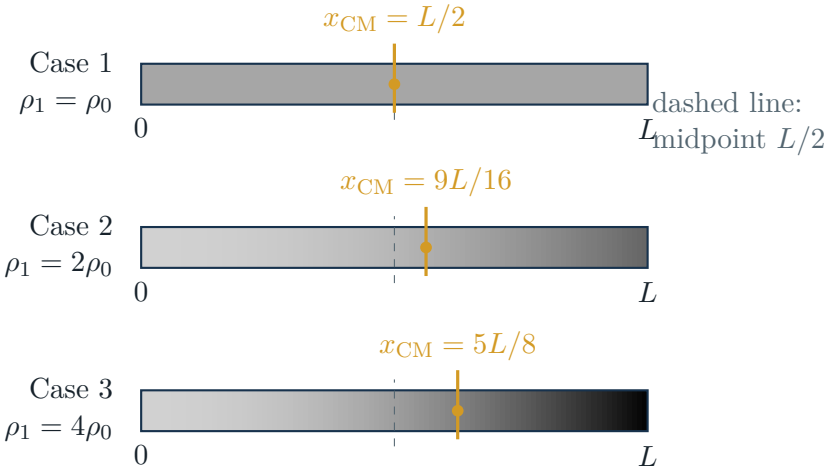
$$x_{\mathrm{CM}} = \frac{3L(\rho_0 + \rho_0)}{4(2\rho_0 + \rho_0)} = \frac{6L\rho_0}{12\rho_0} = \boxed{\frac{L}{2}}.$$

Case 2: $\rho_1 = 2\rho_0$

$$x_{\mathrm{CM}} = \frac{3L(\rho_0 + 2\rho_0)}{4(2\rho_0 + 2\rho_0)} = \frac{9L\rho_0}{16\rho_0} = \boxed{\frac{9L}{16}} = 0.5625L.$$

Case 3: $\rho_1 = 4\rho_0$

$$x_{\mathrm{CM}} = \frac{3L(\rho_0 + 4\rho_0)}{4(2\rho_0 + 4\rho_0)} = \frac{15L\rho_0}{24\rho_0} = \boxed{\frac{5L}{8}} = 0.6250L.$$



As the right end becomes denser, the center of mass moves progressively to the right:

$$\frac{L}{2} = 0.500L \quad \longrightarrow \quad \frac{9L}{16} = 0.5625L \quad \longrightarrow \quad \frac{5L}{8} = 0.6250L.$$

It does not move all the way to the right end because every part of the rod still contributes mass.